

# INTERNAL AND EXTERNAL COMMANDS IN UNIX

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## 1. Internal Commands

**Definition:** Internal commands are built directly into the shell. They are executed by the shell itself without needing to load any separate program from the disk.

**Common Examples:** `cd`, `echo`, `exit`, `read`, `pwd`

```
$ cd /home/indra
$ echo "Hello UNIX"
$ exit
```

## 2. External Commands

**Definition:** External commands are stored as executable files on the disk. When called, the shell searches for them using the PATH variable, loads them into memory, and then executes them.

**Common Examples:** `ls`, `cat`, `cp`, `mv`, `rm`, `grep`, `find`

```
$ ls
$ cat file.txt
$ cp file1.txt file2.txt
```

## Comparison: Internal vs External

| Feature     | Internal Command | External Command |
|-------------|------------------|------------------|
| Stored In   | Shell            | Disk File        |
| Executed By | Shell itself     | Separate Program |
| PATH Search | No               | Yes              |

| Feature         | Internal Command | External Command |
|-----------------|------------------|------------------|
| Execution Speed | Faster           | Slower           |
| Memory Used     | Less             | More             |

**MEMORY TRICK**

Internal Command = **Inside** Shell | External Command = **Outside** Shell (Stored on Disk)

# UNIX FILE TYPES AND PERMISSIONS

## UNIX File Types Summary

| File Type              | Purpose                              | Quick Reference        |
|------------------------|--------------------------------------|------------------------|
| Ordinary File          | Stores user data                     | Data                   |
| Directory File         | Stores filenames and inodes          | Names + Inodes         |
| Character Special File | Character device access (unbuffered) | Keyboard/Terminal      |
| Block Special File     | Block device access (buffered)       | Disk                   |
| FIFO File              | Inter-process communication          | Pipe                   |
| Symbolic Link File     | Points to another file               | Shortcut               |
| Socket File            | Process/network communication        | Communication Endpoint |

## Absolute (Numeric) Permissions

| Number | Binary | Permission | Meaning                |
|--------|--------|------------|------------------------|
| 0      | 000    | ---        | No Permission          |
| 1      | 001    | --x        | Execute Only           |
| 2      | 010    | -w-        | Write Only             |
| 3      | 011    | -wx        | Write + Execute        |
| 4      | 100    | r--        | Read Only              |
| 5      | 101    | r-x        | Read + Execute         |
| 6      | 110    | rw-        | Read + Write           |
| 7      | 111    | rwX        | Read + Write + Execute |